EXPERIMENT 7: API INTEGRATION WITH ZOHO CREATOR

AIM

To integrate Zoho Creator with external applications and services using Application Programming Interfaces (APIs) to enable secure data exchange, automate academic processes, and extend the functionality of the Student Attendance and Academic Performance Tracker.

THEORY

API integration allows applications to communicate and exchange data with external systems. It enables developers to retrieve, create, update, and manage records between applications.

In the Student Attendance and Academic Performance Tracker, APIs can be used to retrieve student details, attendance records, and academic marks through the Zoho Creator REST API. The data can then be analyzed in Zoho Analytics to create dashboards and reports. External services can also be accessed using Deluge's invokeUrl function and configured OAuth connections.

PROCEDURE

1. Log in to Zoho Creator and open the Student Attendance and Academic Performance Tracker application.
2. Identify the external application or service to integrate, such as Zoho Analytics or the Creator REST API.
3. Configure the required API access and authentication settings using an OAuth connection.
4. Configure the necessary API credentials and scopes.
5. Use Deluge scripts or supported integration options to send and receive data.
6. Configure the request URL, HTTP method, headers, and parameters.
7. Execute the API request and verify the response.
8. Validate the returned data and handle errors to ensure reliable operation.

APPLICATIONS OF API INTEGRATION IN THIS PROJECT

- Sending student details to Zoho Analytics for dashboards and reports.
- Retrieving attendance records for attendance analysis.
- Retrieving academic marks for student performance analysis.
- Sending email notifications when attendance or marks fall below a specified threshold.

- Integrating external AI services for academic performance prediction.
- Synchronizing student information with other educational management systems.

REQUEST DETAILS

Item	Value
Method	GET
Target	Student_Details report through the Zoho Creator REST API
Authentication	OAuth connection created in Zoho Creator
Scope	Read access to the required report
Used for	Retrieving student details for reports, dashboards, and external systems

SAMPLE DELUGE SCRIPT: FETCHING STUDENT RECORDS THROUGH AN API CONNECTION

Replace the placeholders with your actual account owner name, application link name, report link name, and connection link name.

```
baseUrl =
"https://www.zohoapis.in/creator/v2.1/data/OWNER_NAME/APP_LINK_NAME/report
/Student_Details";


response = invokeUrl
[
    url : baseUrl
    type : GET
    connection : "creator_connection"
];


info response;


if(response.get("code") == 3000)
{
```

```
    records = response.get("data");
    info "Records fetched: " + records.size();
}
else
{
    info "API error: " + response;
}
```

Note: Configure creator_connection with the required OAuth permissions. Use the correct API domain and endpoint for your Zoho data center, and ensure that OWNER_NAME, APP_LINK_NAME, and Student_Details match the actual API owner, application link name, and report link name. The account must have permission to access the report.

PROCESS DIAGRAM

Figure 1: API Integration Flow

Student Attendance and Academic Performance Tracker
↓
Zoho Creator REST API
↓
OAuth Authentication
↓
Deluge invokeUrl Function
↓
Retrieve Student and Attendance Records
↓
Zoho Analytics
↓
Academic Reports and Dashboards

EXPECTED RESPONSE FORMAT (ILLUSTRATIVE)


```
{
  "code": 3000,
  "data": [
    {
      "Student_ID": "101",
      "Student_Name": "Arun",
```

```
    "Department": "AIML",
    "Year": "3"
   },
   {
    "Student_ID": "102",
    "Student_Name": "Priya",
    "Department": "AIML",
    "Year": "3"
   }
 ]
}
```


The field names and response structure depend on the actual form fields and Creator API configuration.

ERROR HANDLING CHECKLIST

- Check whether the response code is 3000 before processing the data.
- Log the response when an error occurs.
- Verify the API URL, application link name, and report link name.
- Ensure the OAuth connection has the required permissions.
- Never hardcode access tokens or secret credentials in the Deluge script.
- Retrieve only the required records to improve performance.

METRICS BUILT IN ZOHO ANALYTICS FROM SAMPLE DATA

The following values are illustrative expected outputs for the five sample students.

Metric	Value
Total students	5
Students in AIML	5
Students with attendance below 75%	2
Students with final marks below 40	0
Average final mark	73.6

These metrics assume the sample dataset contains the five students listed in the project report. Actual results depend on the records stored in Zoho Creator.

RESULT

Zoho Creator API integration was demonstrated for the Student Attendance and Academic Performance Tracker using a Deluge invokeUrl request and an OAuth connection. The integration enables student records to be retrieved for external reporting and analysis.

CONCLUSION

API integration in Zoho Creator enables communication between applications and external services. In the Student Attendance and Academic Performance Tracker, it supports student data retrieval, attendance analysis, academic reporting, and workflow automation. With appropriate authentication, data validation, and error handling, the application can be extended into a more flexible and connected academic management system.